

How Far Do Threshold Rules Go? Rodent Behavior Classification from Low-Dimensional CPU-Only Tracking Features under Cross-Animal and Cross-Lab Shift

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Abstract—Low-cost rodent trackers run on a laptop CPU, reducing each frame to a few silhouette measurements that hand-set threshold rules turn into behavior labels. We ask what those rules give up against learned classifiers on identical inputs. Using 20 public open-field videos scored by three raters (296k frames) and another laboratory’s recordings as an external test, we compute 14 per-frame features and compare fixed and tuned rules with a decision tree, a random forest, and gradient boosting under one pre-registered protocol. A priori rules almost never fired (macro F1 0.03): seen from above, a rearing mouse barely shortens, so the intuition behind them fails. Tuning within the pre-registered range reached 0.14. A random forest on the same features reached 0.70 against a human pairwise ceiling of 0.79, beating tuned rules in all 20 leave-one-video-out folds; restricted to the five inputs the rules use it still reached 0.70, so the limitation is the rule form, not the features. Under laboratory shift the gap narrowed from 0.57 to 0.43. The pipeline runs at 138 fps on one CPU core.

Index Terms—behavior classification, threshold rules, random forest, distribution shift, animal tracking, open-field test

I. INTRODUCTION

Rearing and self-grooming are standard read-outs of exploration, anxiety, and stereotypy in rodent open-field tests [2], [3], and scoring them by hand costs 40–60 min per 10-min video [13]. Commercial trackers [4], [5] and open pose-based pipelines [8], [10], [13] automate this, but the former are expensive and the latter need GPUs, labeled training frames, and programming skills that many teaching and small laboratories lack. Low-cost trackers therefore reduce each frame to a handful of silhouette features (position, area, body length) that a laptop CPU can compute in real time [6].

Behavior labels from such trackers are almost always produced by hand-written threshold rules, for example “the animal rears when its silhouette shortens.” Rules are transparent and need no training data, but they encode an intuition about what a behavior looks like from above, and we are not aware of a controlled test of that intuition against the obvious alternative: a lightweight learned classifier given *exactly the same* low-dimensional features. If the gap is small, rules are the right tool for low-resource labs; if it is large, a few minutes of CPU training on public labels buys most of the accuracy.

We ask four questions. (RQ1) On identical features, how large is the gap between rules and learned models, and does it differ across supported rearing, unsupported rearing, and grooming? (RQ2) Which approach degrades less when the animal changes (leave-one-video-out) and when the laboratory, camera, and annotators change? (RQ3) How much do temporal context and the source of head information contribute? (RQ4) Does a learned model keep CPU-only processing faster than real time?

Our contributions are: (i) a controlled comparison of fixed and tuned threshold rules against decision-tree, random-forest, and gradient-boosting classifiers on one set of 14 per-frame features, including a random forest restricted to the rules’ own inputs; (ii) evaluation under animal shift (20 videos, 3 raters, 296k frames) and lab shift (an independent laboratory); (iii) ablations of window length and feature source, including a dense-pose reference; and (iv) a CPU cost analysis, with code and per-frame features released and all analysis choices pre-registered before labels were joined to features.

II. RELATED WORK

Rule-based trackers. EthoVision [4], ANY-maze [5], and the open-source ezTrack [6] segment the animal from the background and derive behaviors from thresholds on its trajectory and silhouette; background subtraction with adaptive Gaussian mixtures (MOG2) [7] is the usual CPU-friendly front end.

Pose and learning-based pipelines. DeepLabCut [8] made markerless keypoint tracking practical; supervised classifiers on top of keypoints include JAABA [9], SimBA [10], and the DLCAnalyzer network of Sturman *et al.* [13], which reached human-level rearing accuracy on the dataset we reuse; B-SOiD [11] and the methods compared by Mlost *et al.* [14] discover behaviors without labels, and DeepEthogram [12] classifies raw pixels. All of these assume dense pose or raw video and, in practice, a GPU.

Positioning. We do not compete with these systems [1]. We hold the input fixed at what a CPU silhouette tracker produces and ask whether replacing its rules with a small learned model is worth it.

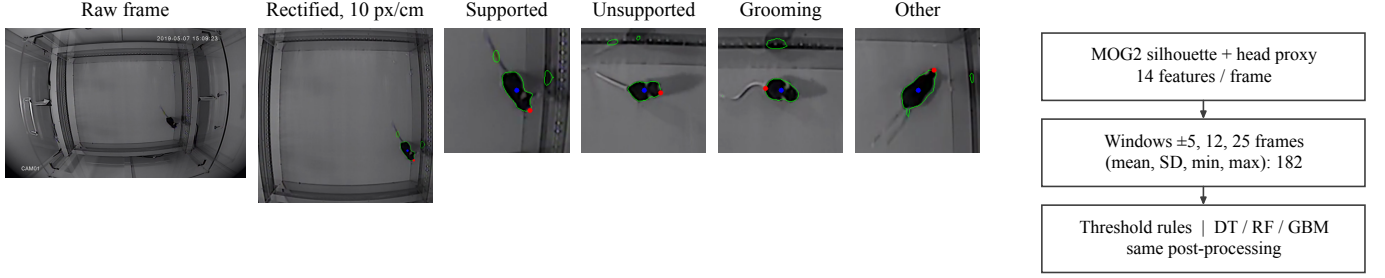


Fig. 1. Pipeline. Frames are rectified to a metric top-down canvas; MOG2 gives a silhouette (green contour, blue centroid), from which 14 features, including a contour head proxy (red), are computed; window statistics feed either threshold rules or a learned classifier, with identical post-processing. Crops show consensus examples from one video.

III. DATA

A. Primary set (ETH). We use the 20 open-field videos of Sturman *et al.* [13] (Zenodo 10.5281/zenodo.3608658): one C57BL/6 mouse per 10-min top-down recording, 928×576 px at 25 fps. Three trained raters independently annotated supported rears, unsupported rears, and grooming as on-set/offset times (DLCAnalyzer repository, label file at commit d6f9532). We converted each rater’s bouts to frames ($\lceil \text{round}(25t_{\text{on}}) \rceil, \lceil \text{round}(25t_{\text{off}}) \rceil$), kept only frames inside the trial markers of every rater who set them (one rater lacked an end marker on two videos), and excluded frames any rater marked as jumping or as a tool placeholder. A rater who assigned two behaviors to the same frame abstained on it. Ground truth is the class chosen by at least two raters; the 0.57% of frames without such a majority were excluded (treating them as “other,” as initially planned, is reported as a sensitivity analysis). This leaves 295,741 scored frames (Table II).

B. External set (Stockholm). The dataset of Mlost *et al.* [14] (Zenodo 10.5281/zenodo.14382973) was recorded in another laboratory with a different camera (650×600 px, 25 fps, 40×40 cm white arena). Three experts labeled 2,999 frames from three mice with ten classes by consensus; we mapped supported rear, unsupported rear, and groom to our classes and the other seven to “other,” as fixed before the test. We matched the label files to videos without using labels, through frame counts and the mutual information between the deposited unsupervised cluster sequences and the pose tracks. Grooming is rare in this set (68 frames).

C. Human agreement. Pairwise frame-level F1 between raters, our human ceiling, was 0.89 (supported), 0.75 (unsupported), and 0.73 (grooming), with Fleiss’ $\kappa = 0.80$.

IV. METHOD

A. Tracker. Fig. 1 summarizes the pipeline. Each frame is warped by a homography onto a top-down canvas of 10 px/cm (ETH: the four arena reference points tracked in the original study, spanning 42×42 cm; Stockholm: floor corners detected automatically on the background image). A MOG2 background model [7] (history 500, variance threshold 16, shadows removed) is first trained on every 50th in-trial

frame and then applied with a frozen background, so that a motionless animal is not absorbed; we then apply a 21×21 Gaussian blur before subtraction, morphological closing, and take the largest contour. The head end of the fitted ellipse is chosen by recent movement direction. Tracking covered $\geq 98.8\%$ of frames in every video, and its centroid agreed with the original study’s pose estimate to a median of 0.5 cm at zero lag.

B. Features. Fourteen per-frame features, all in cm, cm/s, or within-video ratios: speed and acceleration (± 2 -frame differences), area and ellipse major axis relative to the video median, aspect ratio, axis angular speed, solidity, signed centroid-to-wall distance, and motion energy (mean frame difference inside the bounding box, relative to the video median); plus a contour *head proxy* replacing keypoints, which a CPU rodent pose model does not provide: head extent, head-to-wall distance, minimum contour-to-wall distance, head jitter (± 5 frames), and segmentation purity. Learned models additionally see the mean, SD, min, and max of each feature in centred windows of $\pm 5, 12$, and 25 frames (0.2–1 s), giving 183 inputs (182 features and a missing-data flag).

C. Rules. The *fixed* rules were written before seeing data: rear if body-length ratio < 0.75 and area ratio < 0.8 for ≥ 8 frames, supported if the head is within 3 cm of a wall; groom if speed < 2 cm/s, $0.6 < \text{area ratio} < 1.0$, and motion energy above the video’s median for slow frames, for ≥ 25 frames. *Tuned* rules search the same thresholds on the training videos only (staged grid, macro-F1 objective). After seeing the grid saturate, we added a post-hoc *wide* grid (rear thresholds up to 1.10), reported separately.

D. Learned models. A decision tree (depth 6), random forest (RF; 300 trees) [15], and histogram gradient boosting (GBM; 200 iterations) [16], all from scikit-learn [17] with balanced class weights and no hyperparameter tuning. As a control, an RF sees only the five features the rules use (and their windows).

E. Post-processing. Every method’s output passes the same 9-frame majority filter, after which behavior bouts shorter than 5 frames become “other.”

F. Protocol. Leave-one-video-out (LOVO) cross-validation over the 20 ETH videos; for the lab shift, every method is

trained (or tuned) on all 20 and applied once to Stockholm. The headline metric is the macro F1 over the three behaviors; bout counts are compared with the raters’ median per video. The primary test is an exact Wilcoxon signed-rank test over the 20 folds (RF vs. tuned rules), with Holm correction for secondary tests. Thresholds, grids, hyperparameters, and tests were committed to a pre-registration before any label was joined to a feature.

V. RESULTS

A. Animal shift (LOVO). Table I and Fig. 2 summarize the 20 folds. The fixed rules almost never fired (macro F1 0.03). The reason is visible in the features: from a top-down camera a rearing mouse barely shortens, so the median body-length ratio was 0.97 (supported) and 1.00 (unsupported), and only 3% and 5% of rearing frames met the rule’s condition, whereas 29% of grooming frames did, since grooming is the most compact posture (0.77). Tuning helped little: every fold chose the top of the rear length-ratio grid (<0.9 in 20/20 folds), for a macro F1 of 0.14, and even the post-hoc wide grid, which also settled on its edge (1.1), reached only 0.26. The head-to-wall distance, in contrast, separated the two rear types well (median -1.8 cm vs. 4.3 cm).

All learned models were far better. RF reached a macro F1 of 0.70 ± 0.11 and GBM 0.74 ± 0.10 , against a human pairwise ceiling of 0.79; RF beat the pre-registered tuned rules in every fold (median difference 0.58, exact Wilcoxon $p = 1.9 \times 10^{-6}$, rank-biserial $r = 1.00$; all secondary tests $p_{\text{Holm}} \leq 1.3 \times 10^{-5}$). On supported rearing RF came close to the raters (0.89); unsupported rearing (0.61) and grooming (0.61) stayed further below them. Because the pairwise ceiling scores raters against each other and the models against a majority vote, we also compute the matched reference—each rater against the consensus of the other two, models against the same target—on which the raters lead RF by 0.94 vs. 0.92 on supported rearing and 0.87 vs. 0.77 on macro F1. Crucially, an RF given only the five features the rules use scored 0.70, so the rules’ deficit lies in their fixed conjunctive thresholds rather than in the information available to them. Contrary to our hypothesis, the rule-to-RF gap was largest for supported rearing (0.69) rather than unsupported rearing (0.47; difference -0.23, 95% CI [-0.28, -0.17]). Bout counts per video followed the raters’ median closely for RF on supported rears ($r = 0.87$) and less for unsupported rears ($r = 0.76$) and grooming ($r = 0.32$), versus $r = 0.95, 0.96$, and 0.72 between one rater and the other two.

B. Lab shift. Every method was retrained or re-tuned on all 20 ETH videos and applied once to the 2,999 labelled Stockholm frames (Fig. 3a). As predicted, the gap narrowed: RF fell from 0.70 to 0.62 (-0.09) and GBM, the best model in-distribution, fell furthest, from 0.74 to 0.44 (-0.30), whereas the tuned rules rose slightly from their very low base (0.14 to 0.19, +0.05). The RF-minus-rules gap therefore shrank from 0.57 to 0.43. This narrowing reflects the learned models giving up accuracy under shift, not the rules becoming useful: no rule variant exceeded 0.19 macro F1 (0.21 with stand-and-sniff

excluded), and RF remained the best method on the external set. Supported rearing reached the highest external F1 (RF 0.87, 95% CI [0.78, 0.92]), unsupported rearing less well (0.60 [0.43, 0.74]), and grooming, with only 68 positive frames, is not estimable with useful precision (0.38 [0.00, 0.76]); the two-behavior macro F1 is therefore the more informative summary (RF 0.74). Excluding the “stand and sniff” frames, which the source dataset groups with rearing, changes RF to 0.63. With three videos and one consensus annotation these numbers are descriptive only.

C. Ablations. Temporal context mattered: RF on the raw features alone (no windows) scored 0.63, one ± 0.2 s window raised it to 0.66, and the three pre-registered windows gave 0.70 (Fig. 3b). Feature source did not: replacing the five contour head-proxy features by the original study’s DeepLabCut pose features changed the macro F1 from 0.70 to 0.70, and dropping them entirely (nine silhouette features) gave 0.70 (Fig. 3c). Both label sensitivity analyses left the picture unchanged (RF 0.70 with ambiguous frames counted as “other” and 0.71 without the 1-s placeholder bouts; tuned rules 0.14 and 0.14).

D. Compute. On one core of an Apple M4 laptop, a single tracking process ran at 139 fps at 928×576 (44–63 per process when all videos are tracked concurrently) and the features and window statistics added 3 μs per frame. Classification was never the bottleneck: 32.0 μs per frame for RF and 8.2 μs for GBM against 0.1 μs for the rules, so every method ran end to end at 138–139 fps, above the 25 fps capture rate; switching from rules to a forest cost 0.5% of end-to-end throughput. They differ in stored size (55.8 MB for the forest, 1.6 MB for boosting), which matters for shipping a model rather than for running one. Because the windows are centred, the label for a frame is emitted 1 s after it: this is real-time throughput with a fixed 1 s delay, not a zero-latency online system.

E. Error analysis. Pooled over folds, the rules confused the two rear types mainly where the wall geometry is ambiguous: for the tuned rules the supported/unsupported error rate among rear frames was 30% in corners (within 8 cm of two walls) against 19% along a single wall and 0% in the centre, and it peaked when the head sat 0–4 cm from a wall (57–59%). RF reduced the same corner error to 1%.

VI. DISCUSSION AND LIMITATIONS

Three conclusions follow. First, the intuition behind silhouette rules—“rearing shortens the animal”—does not survive a top-down camera: rearing changed the silhouette far less than grooming did, so thresholds written from that intuition fired almost never, and tuning them inside or even outside the pre-registered range did not repair them. Second, the deficit is not a lack of information. An RF restricted to the five features the rules use recovered 0.70 macro F1, within 0.01 of the full RF, so what the rules lack is the ability to combine those same features flexibly rather than through fixed conjunctive cuts. Third, the rules were not uniformly bad: the head-to-wall distance separated supported from unsupported rears cleanly, which is exactly the geometric quantity a rule can express,

TABLE I

LEAVE-ONE-VIDEO-OUT RESULTS ON ETH (FRAME F1, MEAN $\pm$ SD OVER 20 FOLDS), BOUT-COUNT CORRELATION WITH THE RATERS' MEDIAN (PEARSON r , MEAN OVER THE THREE BEHAVIORS), CLASSIFIER COST PER FRAME (BATCH, ONE CORE, EXCLUDING TRACKING AND FEATURES), AND MACRO F1 ON THE EXTERNAL STOCKHOLM SET. $\dagger$ POST-HOC, NOT PRE-REGISTERED.

Method	Supported	Unsupported	Grooming	Macro F1	Bout r	μ s/frame	Stockholm
Rules (fixed)	.03 $\pm$.04	.03 $\pm$.04	.02 $\pm$.09	.03 $\pm$.03	.32	0.1	.12
Rules (tuned)	.19 $\pm$.09	.14 $\pm$.08	.07 $\pm$.09	.14 $\pm$.05	.19	0.1	.19
Rules (tuned, wide) $\dagger$	.48 $\pm$.06	.23 $\pm$.11	.08 $\pm$.12	.26 $\pm$.06	.27	0.1	.12
Decision tree	.81 $\pm$.05	.47 $\pm$.15	.34 $\pm$.15	.54 $\pm$.09	.45	0.2	.47
Random forest	.89 $\pm$.03	.61 $\pm$.13	.61 $\pm$.24	.70 $\pm$.11	.65	32.0	.62
Gradient boosting	.89 $\pm$.03	.66 $\pm$.13	.66 $\pm$.22	.74 $\pm$.10	.65	8.2	.44
RF, rule inputs	.87 $\pm$.03	.60 $\pm$.14	.62 $\pm$.24	.70 $\pm$.11	.70	17.5	.52
Human (pairwise)	.89	.75	.73	.79	.88	—	—

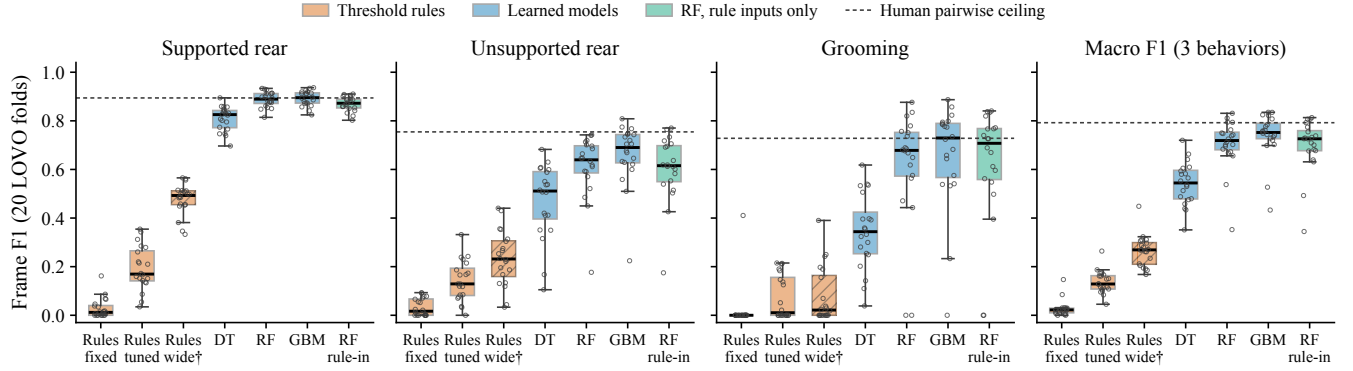


Fig. 2. Per-behavior frame F1 over the 20 LOVO folds. Each open circle is one held-out video; boxes give the median and quartiles and whiskers reach the furthest video within 1.5 IQR. Color marks the method family (legend); the hatched box is the post-hoc wide grid. Dashed line: mean pairwise F1 between the three human raters.

TABLE II

CONSENSUS LABELS. ETH: 20 VIDEOS, 3 RATERS (MAJORITY VOTE);
HUMAN F1 IS THE MEAN PAIRWISE INTER-RATER FRAME F1.
STOCKHOLM: LABELLED FRAMES (3 VIDEOS).

Class	ETH (primary)				Stockh.
	Frames	Bouts	Bout (s)	Human F1	Frames
Supported	48,687 (16.5%)	992	2.0	0.89	312
Unsupported	27,476 (9.3%)	656	1.7	0.75	330
Grooming	7,481 (2.5%)	69	4.3	0.73	68
Other	212,097 (71.7%)	1,712	5.0	—	2289

and it is also where the errors concentrate—in corners for the rules (30%), along single walls for RF (4%).

For a laboratory without a GPU, the practical reading is that a few minutes of CPU training on public labels buys a large, reliable improvement over hand-set thresholds at a cost of 32.0 μ s per frame, while keeping the pipeline interpretable enough to audit. What the learned models did not buy was human-level agreement on the rarer behaviours: unsupported rearing and grooming remained clearly below the human ceiling.

Limitations. One species, one strain, and a single top-down view. Only three behaviors, and no freezing. Grooming is rare (68 labelled frames in the external set, and 1–10 bouts per ETH

video, median 3.5), so its estimates are noisy. Our ground truth converts bout on/offsets to frames, which blurs bout edges. The external set has one consensus annotation rather than independent raters, and its label catalog is not identical to ours, so the mapping of the seven remaining classes to “other” is a judgement call that we fixed before the test and also report with “stand and sniff” excluded. Rectification is exact only on the floor plane, so a rearing body is displaced outward; that is a property of every single-camera top-down setup, not of this tracker alone. Finally, the windows are centred, so an online system would run 1 s behind.

VII. CONCLUSION

On identical low-dimensional CPU features, replacing threshold rules with a small learned classifier changed macro F1 from 0.14 to 0.70 (RF) and 0.74 (GBM), against a human pairwise ceiling of 0.79, and the whole pipeline still runs faster than real time on a laptop CPU. The gap comes from how the rules combine their features, not from what they can see. For resource-limited laboratories that already run a silhouette tracker, training a forest on public labels is a cheap and auditable upgrade; for rearing in particular, a rule written from the top-down “shrinking silhouette” intuition should not be trusted without validation.

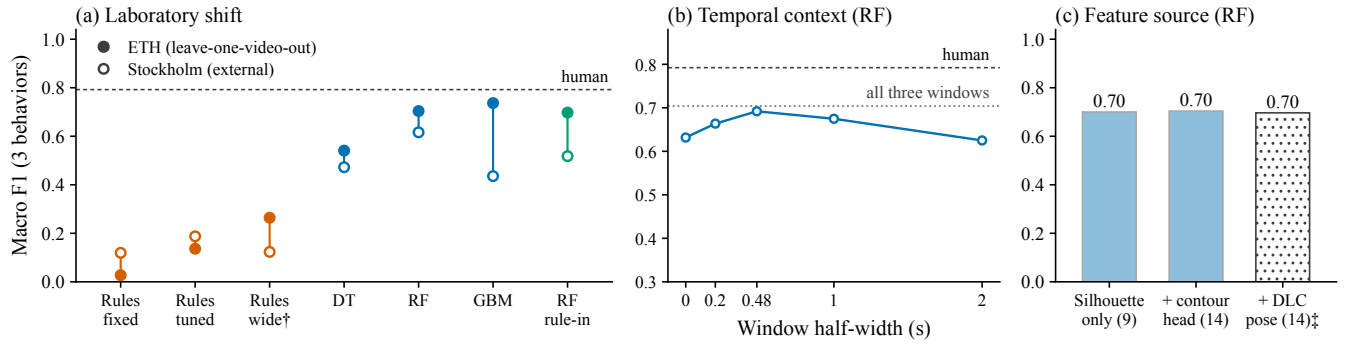


Fig. 3. (a) Macro F1 on ETH (filled markers, mean over the 20 folds) and on the external Stockholm set (open markers), joined per method. (b) RF on the raw features plus a single window of the given half-width; the dotted line is the pre-registered three-window model. Note the zoomed vertical axis in (b); (a) and (c) span the full range. (c) RF by feature source; †DeepLabCut features come from the original study (GPU-trained), shown as a reference rather than as a competing system. Dashed line: mean pairwise F1 between the three human raters.

DATA, CODE, AND ETHICS STATEMENT

No new animal experiments were performed. Videos come from two public CC-BY-4.0 Zenodo records (10.5281/zenodo.3608658; 10.5281/zenodo.14382973); ETH labels and pose files come from the GPL-3.0 DLCAnalyzer repository at pinned commits and are linked, not redistributed. Code, the pre-registration, per-frame features, and predictions are released at <https://github.com/eugenio/rules-vs-learned-rodent-behavior>.

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